

## Comparison Against Current Baseline

| Metric                    | Previous Current Baseline | New Duration-Sensitive Buffer + Aggressive Salvage | Change    |
|---------------------------|---------------------------|----------------------------------------------------|-----------|
| Designed Trips            | 475                       | 475                                                | 0         |
| Scheduled Trips           | 437                       | 441                                                | +4        |
| Unscheduled Trips         | 38                        | 34                                                 | -4        |
| IN Trips                  | 148                       | 150                                                | +2        |
| OUT Trips                 | 227                       | 227                                                | 0         |
| MIXED Trips               | 62                        | 64                                                 | +2        |
| Total Assigned Passengers | 7995                      | 8083                                               | +88       |
| Coverage %                | 92.43%                    | 93.45%                                             | +1.02 pts |
| Avg Occupancy %           | 80.72%                    | 80.89%                                             | +0.17 pts |
| Avg Trip Duration (min)   | 130.36                    | 130.03                                             | -0.33     |
| Total Route Distance (km) | 25765.45                  | 25926.25                                           | +160.80   |
| Max Concurrent Trips      | 12                        | 13                                                 | +1        |
| Fleet Breach vs 13 Buses  | 0                         | 0                                                  | No change |
| Duties Over 9h            | 11                        | 10                                                 | -1        |
| Duties Over 10h           | 0                         | 0                                                  | No change |
| Avg Duty Span (min)       | 352.25                    | 350.86                                             | -1.39     |
| Total Overtime (min)      | 244.31                    | 286.53                                             | +42.22    |
| Total Overtime (hours)    | 4.07 h                    | 4.78 h                                             | +0.70 h   |
| Rescued Trips             | 17                        | 17                                                 | 0         |
| Handover Trips            | 199                       | 203                                                | +4        |

## What Improved

- Coverage improved from 92.43% to 93.45%
- Scheduled trips increased from 437 to 441
- Unscheduled trips fell from 38 to 34
- Assigned passengers increased from 7995 to 8083
- MIXED trips increased from 62 to 64
- Duties over 9h improved from 11 to 10
- Average duty span improved slightly

## What Worsened

- Total overtime increased from 244.31 to 286.53 minutes
- Total route distance increased slightly
- Max concurrent trips moved back to 13, though still without breaching the hard cap

## Why It Improved

The new version improved coverage for two main reasons:

## 1. Duration-sensitive buffer made chaining more realistic

Instead of forcing the same fixed buffer after every trip, the scheduler now uses:

- 15 min after short trips
- 20 min after medium trips
- 30 min after longer trips

This reduced unnecessary `buffer_violation` and allowed more legal trip chaining, especially for short turnaround situations.

## 2. More aggressive fragment salvage recovered leftover demand better

The salvage layer now:

- pools small leftover fragments into broader 60-minute salvage buckets
- retries not only pure `small_isolated_demand`, but also some tiny `buffer_violation` / `slot_exhausted` leftovers

That gave the system better retry trips instead of leaving those fragments stranded immediately.

## Net Interpretation

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This patch successfully improved service coverage under the hard 13 -bus limit while still keeping:

- 0 fleet breaches
- 0 duties over 10h

The tradeoff is a modest overtime increase, but the overall result is directionally positive because the solver now serves more employees and drops fewer trips while staying within the hard operational limits.